Beyond a Single Scale of Intelligence

Looking back on my years in graduate school, I realize how much I learned from the different research styles and personalities of my professors across different areas of maths. From my algebra professors, I learned the importance of working carefully through details, understanding things deeply, finding simple and elegant proofs, and developing a particular perspective on the subject. From my number theory professors, I learned to take a broad and open-minded approach to different ideas and methods, and, even more, to find and formulate questions, and to have the patience to start with small questions and gradually build them up.

In commutative algebra, a subject I care about deeply, one of the most important things I learned from my algebra professor is that sometimes the simplest proofs reflect the deepest understanding. A proof is not only a certificate that a statement is true; the way it is organized can reveal how deeply one sees the underlying structures. This is part of what makes commutative algebra so beautiful to me. Its language can be remarkably simple, but there are hidden structures and rich mathematical phenomena beneath it. There is something deeply aesthetic about the subject to me.

Even though AI now seems to be very good at combining techniques and producing complicated arguments, but I often feel that its mathematical work lacks the personality and individual character that I appreciate in human maths. These are also qualities I learned to recognize through my professors and their different ways of thinking and approaching research, but which I never see from AI. Even with all these developments, I don't feel that the way I approach and enjoy maths needs to change very much.

What AI May Change

AI is becoming increasingly capable of carrying out complicated mathematical proofs, exploring possibilities, and combining existing ideas and techniques. It can discover things that mathematicians have overlooked and provide new results to build on. Recently there have been several AI-generated attempts at major mathematical conjectures, including some that I care about, such as the small Cohen–Macaulay conjecture and Serre's positivity conjecture. We don't know yet whether all of these arguments are correct, and I think they still need serious verification.

I think what makes a conjecture interesting is often the possibility of exploring new mathematical territory, either along the way to a solution or through what the solution itself might reveal. A conjecture usually points toward some underlying structure or theory that we are trying to understand. Sometimes a proof reveals that structure clearly, while other times it settles the conjecture without explaining nearly as much. A counterexample can leave even more questions behind, since it often tells us that something fails without explaining why or under what conditions the conjecture might still hold.

Now what looks good to me in AI-generated maths is its ability to carry out technical arguments and combine existing ideas, rather than its ability to create fundamentally new conceptual viewpoints. This can advance mathematical knowledge, but even when AI generates a proof, further mathematical understanding and insight are often needed to verify it, see what it reveals, and explore how the ideas can be developed further.

Mathematical Insights

Mathematical research involves more than proving difficult statements. Sometimes a mathematician looks at a theorem and feels that it has been formulated in the wrong way. A proof

may be correct but still fail to explain the phenomenon. At other times, one senses that a particular approach is more promising than others, or that existing methods are not enough and a genuinely new mechanism or framework is needed.

Mathematicians often say things like, "There should be a more conceptual explanation," or "There should be a much simpler proof from the right perspective." Such judgments are difficult to formalize. They are not simply reactions to a proof being too long or the hypotheses too complicated; they arise from an internal picture of the subject built through deep mathematical understanding and experience.

That internal picture develops slowly and individually. We may spend years misunderstanding things and pursuing ideas that fail, learning different perspectives, and gradually deepening our understanding. Over time, we develop a personal sense of how a subject behaves, shaped by the problems we have worked on, the mistakes we have made, and the viewpoints that have influenced us. Mathematical intuition is not just a collection of facts, it is also an individual way of organizing them, recognizing what matters, and sensing the structure behind them. Interesting insights often grow out of this kind of deep understanding and experience. There is something profoundly human and individual in this process.

This is another aspect I often find missing in AI-generated mathematical work. Even when the arguments are technically sophisticated and draw on a broad range of ideas, they can feel rather generic, without the individual character or particular insight that one sometimes recognizes in a mathematician's work. At times, an argument may be complicated not because the problem requires it, but because the right perspective is missing. Drawing on a vast amount of existing literature certainly has an advantage but it does not necessarily lead to the most insightful approach.

Different Kinds of Intelligence

AI may become much better than humans at exploring possibilities and carrying out complicated proofs within existing frameworks, while humans may retain distinctive strengths in mathematical taste, conceptual insight, and recognizing which questions and directions are worth pursuing. In mathematical research, these judgments often determine what happens next.

A future mathematician may say to an AI:

"You have proved something much harder than what I could prove myself, but I think you are still looking at the problem in the wrong way."

The mathematician then changes the formulation or introduces a new perspective, and the AI carries out an enormous amount of difficult maths within it.

Maths may develop through collaboration between different kinds of intelligence, but how mathematicians choose to work with AI will also depend on individual preferences. Just as some prefer frequent discussion and collaboration while others work mostly alone, some may find constant interaction with AI productive, while others may use it only occasionally or not at all. Whatever style one chooses, mathematical understanding will remain important, including for judging, modifying, and developing what AI produces.

But I have never believed that maths would cease to matter simply because AI can do certain things better than humans. Mathematics is after all a deeply communal activity. Much of its joy comes not only from solving difficult problems, but also from understanding ideas, appreciating their beauty, and communicating them to others. Different people may find enjoyment in

different aspects of maths, but I do not think our only motivation should depend on the pride of being uniquely capable of doing something. To me, mathematical knowledge and understanding remain interesting regardless of whether they come from a human or a machine.

Of course, AI raises real questions about credit, misuse, education, and responsibility, but much also depends on how people choose to use it. We cannot know exactly how maths will change, but I think the values that make maths meaningful will remain, and may even be deepened as we reconsider the relationship between human and AI.

Maths has never been only about being the best at solving problems. We should continue to enjoy the maths we care about, deepen our understanding, share ideas, and follow the questions that matter to us, becoming the mathematicians we want to be. These are the things that have always mattered, and I believe they will continue to matter both to us individually and to the mathematical community.

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